

MAKE UP EXAMINATIONS - JUNE 2023

Program : B.E :- Computer Science and Engineering
Course Name : Data Structures
Course Code : CS33

Semester : III
Max. Marks : 100
Duration : 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define an abstract data type (ADT). Write an ADT for a complex number and include necessary operations. CO1 (06)
 - How do you perform allocation and deallocation of Memory? Explain using appropriate examples. CO1 (06)
 - Write the pseudocode of the KMP pattern matching algorithm and demonstrate its working using an example. CO1 (08)
- Define what is an algorithm? List the various criteria's to be satisfied by an algorithm. CO1 (05)
 - Write a C function to check equality of Structures. CO1 (05)
 - Illustrate the fast transpose technique to find the transpose of following sparse matrix. CO1 (06)

Row	Col	Value
7	6	8
0	2	45
0	5	7
1	3	8
2	4	18
3	1	10
4	2	32
5	5	15
6	4	-22

- "Pointers can be dangerous", Explain the statement. CO1 (04)

UNIT - II

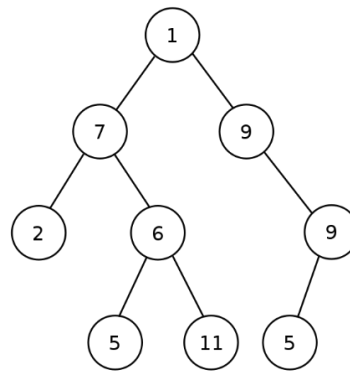
- Write a C functions to perform push and pop operations over a stack implemented using dynamic array. Note: If stack become full double the stack size. CO2 (07)
 - Convert the given infix expression to postfix form using stack. Represent the conversion in tabular form and show.
 $(a+b)*d+e/(f+a*d)+c$ CO2 (07)
 - Compare Linear queue with circular queue with appropriate examples. CO2 (06)
- Write a C functions to perform insert and delete operations over a stack implemented using dynamic array. Note: If queue become full double the queue size. CO2 (07)
 - Evaluate the given postfix expression using stack. Write the steps in tabular form and show.
Note: $a=1, b=2, c=3, d=4, e=5, k=6, g=7, j=8$ $ab+cd/e-kj*+/-$ CO2 (07)
 - Summarize multiple stacks implemented using single array with an example. CO2 (06)

UNIT - III

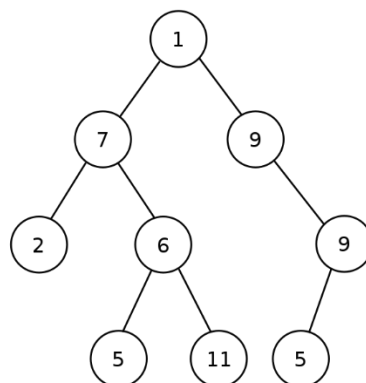
5.
 - a) Write a C function to find out the sum of numbers stored in circular linked list. CO3 (07)
 - b) Outline about the addition of two polynomials represented as circular linked lists with header nodes. Give examples. CO3 (07)
 - c) Write node structure for sparse matrix using linked list. CO3 (06)
6.
 - a) Write a c function to add and delete elements in to multiple queues, implemented using linked lists. CO3 (07)
 - b) Examine the concatenation of two linked lists with different cases using examples. CO3 (07)
 - c) Analyze the doubly linked list insertion and deletion operation along with C functions. CO3 (06)

UNIT- IV

7.
 - a) Make use of two different representations to represent the given Binary tree. Brief about them. CO4 (07)



- b) Show the Satisfiability algorithm with an example propositional formula in a binary tree. Explain in detail. CO4 (07)
 - c) Demonstrate max heap insertion and deletion operations with appropriate examples. CO4 (06)
8.
 - a) Write C functions for inorder(),preorder(),postorder() traversals over binary tree. CO4 (07)
 - b) Draw the threaded linked list representation of the given binary tree . Brief about it. CO4 (07)

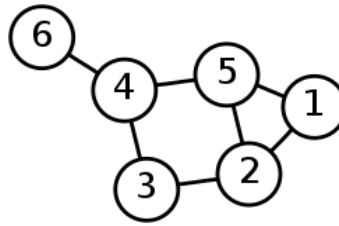


- c) Construct binary tree step by step for the given inorder and preorder traversals. CO4 (06)
 Inorder traversal: 4,10,12,15,18,22,24,25,31,35,44,50,66,70,90
 Preorder traversal : 25,15,10,4,12,22,18,24,50,35,31,44,70,66,90

UNIT - V

9. a) For the given graph

CO5 (06)



Obtain, the following:

- i. The in-degree and out-degree of each vertex.
 - ii. Its adjacency matrix
 - iii. Its adjacency list representation.
- b) What are the different operations performed over a double-ended priority queue? How it is useful in implementing a network buffer? CO5 (06)
- c) Construct a binary tree having the following sequences: CO5 (08)
 Preorder sequence: ACBDFHIG
 Inorder sequence :FEDCABGHI
10. a) Write two C functions to perform a Depth-first search and Breadth-first search of a graph. CO5 (06)
- b) Describe the height-biased leftist trees with suitable examples. CO5 (06)
- c) Consider we have the following key values: CO5 (08)
 7,14,2,5,10,33,56,30,15,25,60,70,4
 Write the AVL tree after each value is inserted into the tree.
